ChE 596 Polymer Rheology - Fall 2024

Homework 2

Due date: Tuesday, September 24 at 1:30 pm EDT

Show all your work for full credit.

Problem 1

State whether true or false. (1 pt each) Explain or justify your answers. (1 pt each)

1. Linear viscoelastic experiments (e.g., dynamic, step-strain) are unable to provide nonlinear rheological properties of materials (e.g., shear thinning).

2. The first normal stress difference N1 is typically a positive number for almost all polymers. [you must state the source of your answer for credit]

3. For a sample with yield stress, below the yield stress the viscosity $\eta \approx$ 10 million Pa.s.

4. The entanglement molecular weight (M_e) of a polymer is independent of temperature.

5. Both $\underline{a}\bullet\underline{\underline{T}}\bullet\underline{b}$ and $\underline{b}\bullet\underline{\underline{T}}\bullet\underline{a}$ are scalars and equal to each other, given that $\underline{a}$ and $\underline{b}$ are vectors and $\underline{\underline{T}}$ is a tensor.

6. For low strain rates, one can predict the extensional viscosity ($\bar{\eta}^{+}$) versus time curve from the shear viscosity (η^{+}) versus time curve.

7. The Hershel-Bulkley model can be used to fit viscosity data of a polymer melt.

Problem 2

You have been asked to characterize the steady state viscosity versus shear rate behavior of a polymer. Unfortunately, you cannot do such an experiment because the sample is too viscous, so it overloads the transducer and comes out of the cone (or plate) and plate gap. Suggest two other experiments you could do that would provide the same information as viscosity versus shear rate. Explain how the material functions of these experiments relate to steady state viscosity.

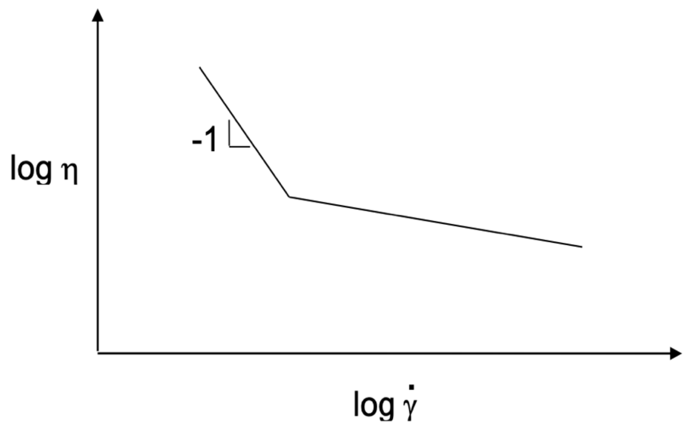
Problem 3

You are the only rheologist in your company and the product division wants elastic modulus (G') data for a polymer as a function of frequency. They are particularly interested in the frequency range between 0.001 and 10^5 rad/s at T=60°C. Your instrument only goes between 0.01 and 100 rad/s and you have no access to other machines. Explain how you can get this data without resorting to other instruments. Use diagrams or sketches to help illustrate your point.

Problem 4

The viscosity - shear rate plot of a material is given below.

(a) Sketch the viscosity versus shear stress behavior of this material.
(b) What kind of material do you think this is and what model could best fit the data? Give reasons.


log η
-1
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